

PRACTICAL - 2

OBJECTIVE:

The objective of this practical is to learn how to install and configure Hadoop in a Single-Node Cluster environment. It also helps in understanding HDFS operations such as storing, managing, retrieving, and monitoring big data efficiently.

AIM: Hadoop Installation, Configuration and HDFS Operations on Windows/Linux

Problem Definition

You have recently joined DataAnalytics Inc. as a Hadoop Administrator.

The organization processes large-scale customer, sales, and marketing data using the Hadoop ecosystem. Due to increasing data volumes, the company has decided to migrate its data storage infrastructure from traditional file systems to Hadoop Distributed File System (HDFS). As a Hadoop Administrator, your responsibility is to install and configure

Hadoop in a Single-Node Cluster environment on Windows/Linux, verify the Hadoop services, manage HDFS storage, and perform data management operations.

Management wants a fully functional Hadoop environment that can store, retrieve, and process big data efficiently.

Your task is to install Hadoop, configure the Hadoop ecosystem, start Hadoop services, and perform HDFS operations using Hadoop commands.

Task:

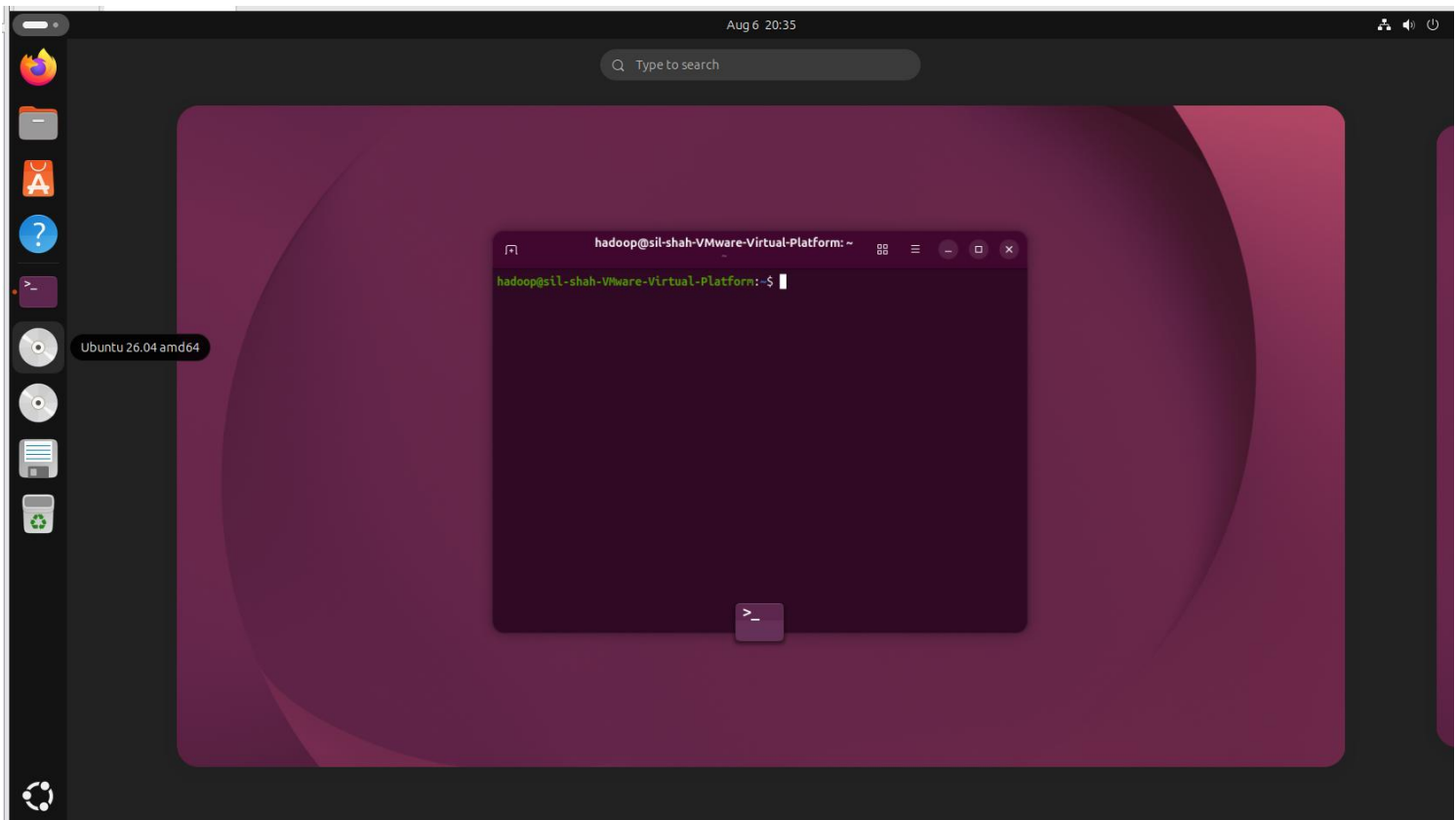
- Hadoop Installation & Configuration
- Access Hadoop Web UI.
- 1. Create directories in HDFS.
- 2. Upload files to HDFS.
- 3. List HDFS files and directories.
- 4. Display file contents.
- 5. Copy files within HDFS.

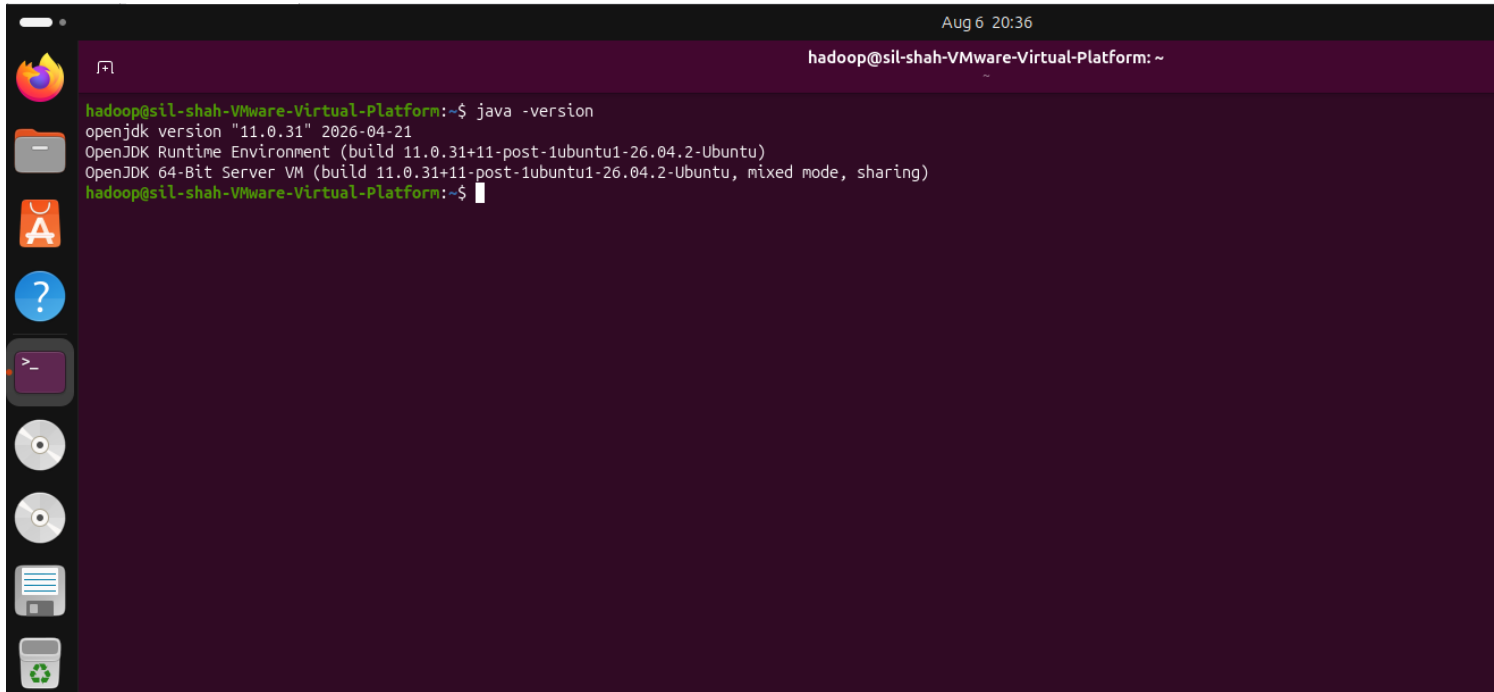
6. Move files within HDFS.
7. Rename files.
8. Download files from HDFS.
9. Delete files and directories.
10. Check HDFS report.
11. Verify block information.
12. Check Hadoop cluster health.
13. Monitor Hadoop processes.
14. Check storage utilization.

Tools / Technology Used

- Windows 10 / Windows 11
- OpenJDK 11 or above
- Hadoop 3.x
- Winutils.exe
- Command Prompt / PowerShell

Step 1: Open Terminal



Step 2: Check JavaA terminal window with a dark purple background. The title bar at the top right says "Aug 6 20:36" and "hadoop@sil-shah-VMware-Virtual-Platform: ~". On the left, there is a vertical dock with icons for a web browser, a file manager, a terminal, and a trash can. The terminal text shows the command "java -version" being executed, resulting in the following output: "openjdk version '11.0.31' 2026-04-21", "OpenJDK Runtime Environment (build 11.0.31+11-post-1ubuntu1-26.04.2-Ubuntu)", and "OpenJDK 64-Bit Server VM (build 11.0.31+11-post-1ubuntu1-26.04.2-Ubuntu, mixed mode, sharing)".

```
hadoop@sil-shah-VMware-Virtual-Platform:~$ java -version
openjdk version "11.0.31" 2026-04-21
OpenJDK Runtime Environment (build 11.0.31+11-post-1ubuntu1-26.04.2-Ubuntu)
OpenJDK 64-Bit Server VM (build 11.0.31+11-post-1ubuntu1-26.04.2-Ubuntu, mixed mode, sharing)
hadoop@sil-shah-VMware-Virtual-Platform:~$
```

Step 3: Check Hadoop VersionA terminal window with a dark purple background. The title bar at the top right says "Aug 6 20:37" and "hadoop@sil-shah-VMware-Virtual-Platform: ~". The terminal text shows the command "java -version" being executed, resulting in the same output as in Step 2. Then, the command "hadoop version" is executed, resulting in the following output: "Hadoop 3.3.6", "Source code repository https://github.com/apache/hadoop.git -r 1be78238728da9266a4f88195058f08fd012bf9c", "Compiled by ubuntu on 2023-06-18T08:22Z", "Compiled on platform linux-x86_64", "Helped with protoc 3.7.1", "From source with checksum 5652179ad55f76cb287d9c633bb53bbd", and "This command was run using /usr/local/hadoop/share/hadoop/common/hadoop-common-3.3.6.jar".

```
hadoop@sil-shah-VMware-Virtual-Platform:~$ java -version
openjdk version "11.0.31" 2026-04-21
OpenJDK Runtime Environment (build 11.0.31+11-post-1ubuntu1-26.04.2-Ubuntu)
OpenJDK 64-Bit Server VM (build 11.0.31+11-post-1ubuntu1-26.04.2-Ubuntu, mixed mode, sharing)
hadoop@sil-shah-VMware-Virtual-Platform:~$ hadoop version
Hadoop 3.3.6
Source code repository https://github.com/apache/hadoop.git -r 1be78238728da9266a4f88195058f08fd012bf9c
Compiled by ubuntu on 2023-06-18T08:22Z
Compiled on platform linux-x86_64
Helped with protoc 3.7.1
From source with checksum 5652179ad55f76cb287d9c633bb53bbd
This command was run using /usr/local/hadoop/share/hadoop/common/hadoop-common-3.3.6.jar
hadoop@sil-shah-VMware-Virtual-Platform:~$
```

Step 4: Start Hadoop

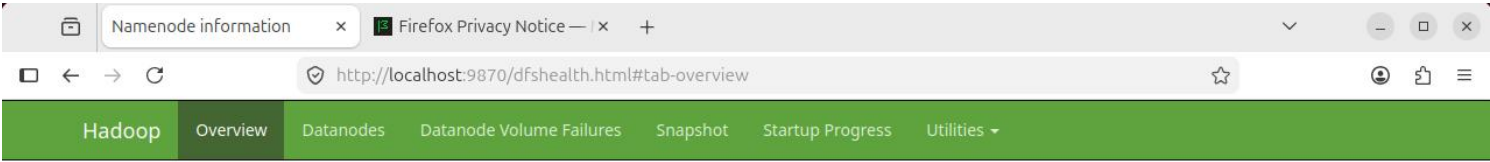
```
Aug 6 20:39
hadoop@sil-shah-VMware-Virtual-Platform: ~

hadoop@sil-shah-VMware-Virtual-Platform:~$ java -version
openjdk version "11.0.31" 2026-04-21
OpenJDK Runtime Environment (build 11.0.31+11-post-1ubuntu1-26.04.2-Ubuntu)
OpenJDK 64-Bit Server VM (build 11.0.31+11-post-1ubuntu1-26.04.2-Ubuntu, mixed mode, sharing)
hadoop@sil-shah-VMware-Virtual-Platform:~$ hadoop version
Hadoop 3.3.6
Source code repository https://github.com/apache/hadoop.git -r 1be78238728da9266a4f88195058f08fd012bf9c
Compiled by ubuntu on 2023-06-18T08:22Z
Compiled on platform linux-x86_64
Compiled with protoc 3.7.1
From source with checksum 5652179ad55f76cb287d9c633bb53bbd
This command was run using /usr/local/hadoop/share/hadoop/common/hadoop-common-3.3.6.jar
hadoop@sil-shah-VMware-Virtual-Platform:~$ start-dfs.sh
Starting namenodes on [localhost]
Starting datanodes
Starting secondary namenodes [sil-shah-VMware-Virtual-Platform]
hadoop@sil-shah-VMware-Virtual-Platform:~$ start-yarn.sh
Starting resourcemanager
Starting nodemanagers
```

Step 5: Verify Services

```
hadoop@sil-shah-VMware-Virtual-Platform:~$ jps
7203 NodeManager
7589 Jps
6837 SecondaryNameNode
6453 NameNode
6602 DataNode
7069 ResourceManager
```

Access Hadoop Web UI



Overview 'localhost:9000' (✓active)

Started:	Thu Aug 06 20:38:34 +0530 2026
Version:	3.3.6, r1be78238728da9266a4f88195058f08fd012bf9c
Compiled:	Sun Jun 18 13:52:00 +0530 2023 by ubuntu from (HEAD detached at release-3.3.6-RC1)
Cluster ID:	CID-d1f05f54-6408-4f6a-bf0a-fc08e21c6ff1
Block Pool ID:	BP-1796484735-127.0.1.1-1784382381912

Summary

Security is off.
Safemode is off.
3 files and directories, 1 blocks (1 replicated blocks, 0 erasure coded block groups) = 4 total filesystem object(s).
Heap Memory used 89.14 MB of 135 MB Heap Memory. Max Heap Memory is 1.31 GB.
Non Heap Memory used 52.34 MB of 55.19 MB Committed Non Heap Memory. Max Non Heap Memory is <unbounded>.

Configured Capacity:	29.36 GB
Configured Remote Capacity:	0 B
DFS Used:	48 KB (0%)

Step 6: Create Dataset

```
hadoop@sil-shah-VMware-Virtual-Platform:~$ nano sales_data.txt
hadoop@sil-shah-VMware-Virtual-Platform:~$ cat sales_data.txt
ProductA 120
ProductB 80
ProductC 150
ProductA 200
ProductB 90
```

Practical Commands

1 Create Directory

```
hadoop@sil-shah-VMware-Virtual-Platform:~$ hdfs dfs -mkdir /sales
hadoop@sil-shah-VMware-Virtual-Platform:~$ hdfs dfs -ls /
Found 2 items
drwxr-xr-x  - hadoop supergroup          0 2026-08-06 20:44 /sales
drwxr-xr-x  - hadoop supergroup          0 2026-07-18 19:18 /test
```

2 Upload File

```
hadoop@sil-shah-VMware-Virtual-Platform:~$ hdfs dfs -put sales_data.txt /sales
```

3 List Files

```
hadoop@sil-shah-VMware-Virtual-Platform:~$ hdfs dfs -ls /sales
Found 1 items
-rw-r--r--  1 hadoop supergroup          63 2026-08-06 20:44 /sales/sales_data.txt
```

4 Display File

```
hadoop@sil-shah-VMware-Virtual-Platform:~$ hdfs dfs -cat /sales/sales_data.txt
ProductA 120
ProductB 80
ProductC 150
ProductA 200
ProductB 90
```

5 Copy File

```
hadoop@sil-shah-VMware-Virtual-Platform:~$ hdfs dfs -cp /sales/sales_data.txt /sales/sales_copy.txt
hadoop@sil-shah-VMware-Virtual-Platform:~$ hdfs dfs -ls /sales
Found 2 items
-rw-r--r-- 1 hadoop supergroup 63 2026-08-06 20:46 /sales/sales_copy.txt
-rw-r--r-- 1 hadoop supergroup 63 2026-08-06 20:44 /sales/sales_data.txt
```

6 Move File

```
hadoop@sil-shah-VMware-Virtual-Platform:~$ hdfs dfs -mkdir /backup
hadoop@sil-shah-VMware-Virtual-Platform:~$ hdfs dfs -mv /sales/sales_copy.txt /backup/
hadoop@sil-shah-VMware-Virtual-Platform:~$ hdfs dfs -ls /backup
Found 1 items
-rw-r--r-- 1 hadoop supergroup 63 2026-08-06 20:46 /backup/sales_copy.txt
```

7 Rename File

```
hadoop@sil-shah-VMware-Virtual-Platform:~$ hdfs dfs -mv /sales/sales_data.txt /sales/customer_sales.txt
hadoop@sil-shah-VMware-Virtual-Platform:~$ hdfs dfs -ls /sales
Found 1 items
-rw-r--r-- 1 hadoop supergroup 63 2026-08-06 20:44 /sales/customer_sales.txt
```

8 Download File

```
hadoop@sil-shah-VMware-Virtual-Platform:~$ hdfs dfs -get /sales/customer_sales.txt
hadoop@sil-shah-VMware-Virtual-Platform:~$ ls
Desktop Documents Downloads Music Pictures Public Templates Videos customer_sales.txt hadoop-3.3.6.tar.gz sales_data.txt snap
```

9 Delete File

```
hadoop@sil-shah-VMware-Virtual-Platform:~$ hdfs dfs -rm /sales/customer_sales.txt
Deleted /sales/customer_sales.txt
hadoop@sil-shah-VMware-Virtual-Platform:~$ hdfs dfs -rm -r /sales
Deleted /sales
```

10 HDFS Report

```
hadoop@sil-shah-VMware-Virtual-Platform:~$ hdfs dfsadmin -report
Configured Capacity: 31523246080 (29.36 GB)
Present Capacity: 15466229686 (14.40 GB)
DFS Remaining: 15466164224 (14.40 GB)
DFS Used: 65462 (63.93 KB)
DFS Used%: 0.00%
Replicated Blocks:
    Under replicated blocks: 0
    Blocks with corrupt replicas: 0
    Missing blocks: 0
    Missing blocks (with replication factor 1): 0
    Low redundancy blocks with highest priority to recover: 0
    Pending deletion blocks: 0
Erasure Coded Block Groups:
    Low redundancy block groups: 0
    Block groups with corrupt internal blocks: 0
    Missing block groups: 0
    Low redundancy blocks with highest priority to recover: 0
    Pending deletion blocks: 0

-----
Live datanodes (1):

Name: 127.0.0.1:9866 (localhost)
Hostname: sil-shah-VMware-Virtual-Platform
Decommission Status : Normal
Configured Capacity: 31523246080 (29.36 GB)
DFS Used: 65462 (63.93 KB)
Non DFS Used: 14429786186 (13.44 GB)
DFS Remaining: 15466164224 (14.40 GB)
DFS Used%: 0.00%
DFS Remaining%: 49.06%
Configured Cache Capacity: 0 (0 B)
Cache Used: 0 (0 B)
Cache Remaining: 0 (0 B)
Cache Used%: 100.00%
Cache Remaining%: 0.00%
Xceivers: 0
Last contact: Thu Aug 06 20:50:37 IST 2026
Last Block Report: Thu Aug 06 20:38:40 IST 2026
Num of Blocks: 2
```

11 Verify Block Information

```
hadoop@sil-shah-VMware-Virtual-Platform:~$ hdfs fsck /
Connecting to namenode via http://localhost:9870/fsck?ugi=hadoop&path=%2F
FSCK started by hadoop (auth:SIMPLE) from /127.0.0.1 for path / at Thu Aug 06 20:51:28 IST 2026

Status: HEALTHY
  Number of data-nodes: 1
  Number of racks:      1
  Total dirs:           3
  Total symlinks:       0
Terminal
Replicated Blocks:
  Total size:    76 B
  Total files:   2
  Total blocks (validated): 2 (avg. block size 38 B)
  Minimally replicated blocks: 2 (100.0 %)
  Over-replicated blocks: 0 (0.0 %)
  Under-replicated blocks: 0 (0.0 %)
  Mis-replicated blocks: 0 (0.0 %)
  Default replication factor: 1
  Average block replication: 1.0
  Missing blocks: 0
  Corrupt blocks: 0
  Missing replicas: 0 (0.0 %)
  Blocks queued for replication: 0

Erasure Coded Block Groups:
  Total size:    0 B
  Total files:   0
  Total block groups (validated): 0
  Minimally erasure-coded block groups: 0
  Over-erasure-coded block groups: 0
  Under-erasure-coded block groups: 0
  Unsatisfactory placement block groups: 0
  Average block group size: 0.0
  Missing block groups: 0
  Corrupt block groups: 0
  Missing internal blocks: 0
  Blocks queued for replication: 0
FSCK ended at Thu Aug 06 20:51:28 IST 2026 in 5 milliseconds

The filesystem under path '/' is HEALTHY
```

12 Check Cluster Health

```
hadoop@sil-shah-VMware-Virtual-Platform:~$ hdfs dfsadmin -report
Configured Capacity: 31523246080 (29.36 GB)
Present Capacity: 15466069942 (14.40 GB)
DFS Remaining: 15466004480 (14.40 GB)
DFS Used: 65462 (63.93 KB)
DFS Used%: 0.00%
Replicated Blocks:
    Under replicated blocks: 0
    Blocks with corrupt replicas: 0
    Missing blocks: 0
    Missing blocks (with replication factor 1): 0
    Low redundancy blocks with highest priority to recover: 0
    Pending deletion blocks: 0
Erasure Coded Block Groups:
    Low redundancy block groups: 0
    Block groups with corrupt internal blocks: 0
    Missing block groups: 0
    Low redundancy blocks with highest priority to recover: 0
    Pending deletion blocks: 0

-----
Live datanodes (1):

Name: 127.0.0.1:9866 (localhost)
Trash: sil-shah-VMware-Virtual-Platform
Decommission Status : Normal
Configured Capacity: 31523246080 (29.36 GB)
DFS Used: 65462 (63.93 KB)
Non DFS Used: 14429945930 (13.44 GB)
DFS Remaining: 15466004480 (14.40 GB)
DFS Used%: 0.00%
DFS Remaining%: 49.06%
Configured Cache Capacity: 0 (0 B)
Cache Used: 0 (0 B)
Cache Remaining: 0 (0 B)
Cache Used%: 100.00%
Cache Remaining%: 0.00%
Xceivers: 0
Last contact: Thu Aug 06 20:52:07 IST 2026
Last Block Report: Thu Aug 06 20:38:40 IST 2026
Num of Blocks: 2
```

13 Monitor Hadoop Processes

```
hadoop@sil-shah-VMware-Virtual-Platform:~$ jps
7203 NodeManager
6837 SecondaryNameNode
6453 NameNode
10185 Jps
6602 DataNode
7069 ResourceManager
```

14 Storage Utilization

```
hadoop@sil-shah-VMware-Virtual-Platform:~$ hdfs dfs -df -h
Filesystem          Size      Used Available  Use%
hdfs://localhost:9000 29.4 G   63.9 K    14.4 G    0%
```

15 Stop Hadoop

```
hadoop@sil-shah-VMware-Virtual-Platform:~$ stop-yarn.sh
Stopping nodemanagers
Stopping resourcemanager
hadoop@sil-shah-VMware-Virtual-Platform:~$ stop-dfs.sh
Stopping namenodes on [localhost]
Stopping datanodes
Stopping secondary namenodes [sil-shah-VMware-Virtual-Platform]
```

Key Questions / Analysis

Q1. Why is Hadoop preferred for Big Data storage?

Hadoop can store and process very large amounts of data across multiple computers. It is fast, reliable, and cost-effective.

Q2. What is the role of NameNode and DataNode in HDFS?

The NameNode manages file information and keeps track of where data is stored. The DataNode stores the actual data blocks and sends them when requested.

Q3. What is the difference between HDFS and traditional file systems?

HDFS stores data across multiple machines, while a traditional file system stores data on a single computer. HDFS is designed for handle.

Q4. What are the advantages of distributed storage?

Distributed storage improves speed, reliability, scalability, and allows data to remain available even if one machine fails.

Supplementary Questions

1. Configure Hadoop pseudo-distributed mode.

Pseudo-distributed mode allows Hadoop services to run on a single computer for learning, testing, and development.

2. Change HDFS replication factor and observe results.

Changing the replication factor changes the number of copies of each data block stored in HDFS, improving data availability and fault tolerance.

3. Upload a large dataset into HDFS.

A large dataset is uploaded into HDFS using Hadoop commands so it can be stored and processed efficiently.

4. Analyze HDFS storage utilization.

Storage utilization is checked to understand how much HDFS space is used and how much is available.

Conclusion

In this practical, Hadoop was successfully installed and configured in a Single-Node Cluster environment. HDFS operations such as creating directories, uploading, downloading, moving, renaming, and deleting files were performed successfully. Hadoop services and storage utilization were also monitored. This practical helped in understanding how Hadoop manages and stores large amounts of data efficiently using HDFS.

Key Skills to be Addressed

- Hadoop Installation and Configuration
- Environment Variable Configuration
- HDFS Administration
- Distributed Storage Management
- Hadoop Cluster Monitoring

Dataset / TestData

sales_data.txt

ProductA 120

ProductB 80

ProductC 150

ProductA 200

ProductB 90

Format

TXT/CSV Files (mb/gb)